REAL ESTATE PRICE PREDICTION

A Regression Analysis of King County, Washington (USA) House Sales

Data Science Capstone

Malak Adel - Cohort: DSB2

Data source: King County House Sales (Kaggle)

Real Estate Price Prediction Using Machine Learning

- Predict house prices based on property characteristics
- Identify the key factors affecting house prices
- Develop a high-performing Stacking Ensemble Model

Problem Statement

- House prices vary depending on property size, quality, location, and other characteristics.
- This makes accurate price estimation difficult for buyers and sellers.
- The goal is to build a model that can provide reliable house price predictions and identify the main price drivers.

Data Summary

- Dataset: King County House Sales
- Records: 21,613
- Features: 25 original features
- Target: House sale price
- Includes property size, bedrooms, bathrooms, grade, condition, location, view, waterfront, and year built

Key Findings

Living area had the strongest relationship with price (r = 0.70)

Grade was also strongly related to price (r = 0.67)

Other important factors included bathrooms, view, basement size, and bedrooms

House prices were highly right-skewed, so a log transformation was tested

Data Cleaning & Feature Engineering

- No missing values were found
- Corrected an unrealistic bedrooms = 33 data entry
- Created house_age
- Created is_renovated
- Created is_studio

MODELING APPROACH

Regression Problem

Models tested:

- **Linear Regression — Baseline**
- **Random Forest**
- **XGBoost**
- **LightGBM**
- **CatBoost**
- **Stacking Ensemble — Final Model**

Evaluation Metrics:

- **MAPE**
- **R^2**

FROM INDIVIDUAL MODELS TO STACKING

Why Stacking?
Individual models capture different patterns in the data
XGBoost: strong predictive performance
LightGBM: efficient gradient boosting
CatBoost: captures additional non-linear patterns
Stacking combines their predictions to improve overall performance
Goal: Combine the strengths of multiple models instead of relying on one mode

STACKING MODEL ARCHITECTURE

Base Models
XGBoost
↓
LightGBM
↓
CatBoost
↓
RidgeCV Meta-Model
↓
Final House Price Prediction

Final House Price Prediction

- **Preprocessing is applied through a pipeline**
- **Predictions from the base models are combined by the final estimator**
- **Final predictions are transformed back to the original price scale**

Model Comparison

Stacking achieved the best overall performance.

Model	R2 Score	(%) MAPE	Notes / Best Parameters
Stacking (Best)	0.914	11.53%	Ensemble Meta-Model
CatBoost	0.909	11.76%	Tuned
XGBoost	0.908	11.65%	Tuned
LightGBM	0.907	11.85%	Tuned
Random Forest	0.872	12.76%	Baseline Tree
Lasso	0.587	15.29%	alpha = 0.0001
Ridge	0.585	15.29%	alpha = 10.0

Important Features

Main factors influencing house prices:

- Grade
- Living Area
- Location
- Latitude
- Zipcode
- View
- Waterfront

These features indicate that property quality, size, and location are major drivers of house price

Limitations & Future Improvements

Lmitations:

- Dataset covers a specific market and time period.
- Some economic and neighborhood factors are not captured.
- Predictions may not generalize to other markets.

Future Improvements:

- Use newer and larger datasets.
- Add more detailed location and market variables.
- Further tune the Stacking Model.
- Test the model on other real estate markets.

Conclusion

Final Model: Stacking Ensemble

- **Combines XGBoost, LightGBM, and CatBoost**
- **Uses RidgeCV as the final estimator**
- **Achieved MAPE = 11.53%**
- **Achieved R^2 = 0.9140**
- **Best-performing model in the project**

Key takeaway:

Combining multiple strong models improved prediction performance and provided a more reliable house price estimation approach.

Thank You

Questions & Discussion

Real Estate Price Prediction Model

DATA SCIENCE CAPSTONE PROJECT | **Malak Adel** (Cohort: DSB2)

An automated machine learning pipeline to deliver accurate, data-driven home valuations for buyers and sellers.

VISUAL PIPELINE FLOW

1 — Data Ingestion

- 21,613 property records
- 25 initial features

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2 — Preprocessing

- Outlier & skewness correction
- Log price transformation
- Engineered house_age & is_renovated

→

3 — Stacking Architecture

- Base models: XGBoost + LightGBM + CatBoost
- Meta-learner: RidgeCV

→

4 — Output

- Optimized house price prediction

KEY PERFORMANCE HIGHLIGHTS

0.914
R^2 Score
Best Variance Explained

11.53%
MAPE
Low Average Error

21K+
Sales Records
Analyzed

KEY PRICE DRIVERS

Living Area (Sqft)
Strongest Impact (r = 0.70)

Building Grade
Quality & Design (r = 0.67)

Location
Latitude & Zipcode Insights